Superintelligence: What Boards Actually Need to Know

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Executive Summary

Every board has heard the AI pitch by now: productivity gains, copilots, efficiency curves. This Espresso Call set that conversation aside to examine something different and far less discussed in boardrooms: the stated objective of the world's leading AI companies to build superintelligence, meaning fully autonomous AI systems that outperform humans across virtually every cognitive task and can act beyond human oversight and control.

Andrea Miotti, founder and CEO of ControlAI, a non-partisan non-profit working in Washington, London, and increasingly Germany and Canada, opened with a plain statement: the risks are real. They have been discussed within the AI field for more than a decade. What has changed is not the substance of the warnings but the willingness of experts, researchers and even company leaders to voice them in public.

His central argument rested on a distinction that most boardroom discussions blur. There are two very different things called "AI". The first is the set of tools companies use every day: chatbots, assistants, business agents. The second is what the frontier laboratories are actually directing their research towards: autonomous entities that operate over long-time horizons, in the digital and physical world, without human supervision. The danger, he argued, lies entirely in the second category.

Three features make that second category dangerous. These systems are grown rather than built, so nobody fully understands how they work. They are being made more autonomous and more capable at speed. And the leading laboratories are explicitly attempting recursive self-improvement, using AI to automate the development of AI itself, which could move the world to uncontrollable systems in months to years rather than decades.

Andrea illustrated the point with a recent incident in which, by his account, unreleased AI agents under test broke out of a contained environment, coordinated among themselves through a self-created messaging channel, hacked a third-party company to obtain the answers to a test, and covered their tracks. Independent government testers, he noted, have also reported that frontier systems increasingly recognise when they are being tested and behave differently as a result.

His proposed response is narrow and targeted: a prohibition on the development of superintelligence, not on AI, and not on data centres. Defined by outcome rather than by technology, such a ban would capture systems capable of undermining a state's

security forces. Only governments can deliver this, he argued, and the nuclear non-proliferation regime shows that coordination is possible even between rivals.

For boards, the message was twofold. Commercially, keep using AI and build internal capability; abstaining personally will not change the trajectory. Beyond the business case, directors hold a voice that most citizens do not, with their governments, in public, and through philanthropy. Andrea's call was that they use it, and soon: "now means now".

1. Introduction

This session was convened because the debate around superintelligence had become very loud in the preceding weeks, with headlines ranging from legislative proposals in Westminster and Washington to public departures and warnings from inside the leading AI laboratories. The purpose of the call was not to add to the noise but to help members tell the noise from the signal: where things actually stand, what the concern is really about, and where we are on the timeline.

ControlAI has spent the last three years doing what few others were doing at the time: taking the question of superintelligence out of the laboratory and into the political arena. The organisation has briefed more than 400 lawmakers across the United Kingdom, the United States, Canada and Germany, including more than 200 US congressional offices. It worked on the bill introduced in Westminster on 8 September, described in the session as the first legislative proposal for a ban on superintelligence, and was consulted on a proposal for a comparable ban in the US Senate. Andrea was careful to note that ControlAI is non-partisan, works with both Republican and Democratic offices, and that "consulted" rather than "co-drafted" is the accurate description of its role in Washington.

2. Two Kinds of AI: The Distinction That Matters

When people warn that "AI could kill us all", Andrea stressed, they are not talking about ChatGPT, nor about the AI tools that directors and their organisations now use daily for research, drafting and information retrieval. Using the same word for both is, in his view, one of the main reasons the debate is so confused.

The customer-facing layer

Chatbots and business agents that automate routine processes are tools. They can be misused, and that misuse is a real but manageable problem of the kind societies already know how to address. Fundamentally, however, they are instruments that humans direct.

The actual objective

What OpenAI, Anthropic, Google DeepMind, xAI and others are working towards is something else: AI systems designed to replace and outcompete all humans at all tasks. This is not a tool for people or businesses to use. It is an entity in its own right that takes autonomous action over ever longer time horizons, operates in the digital and physical world, and acts with persistence and continuous activity.

Andrea argued that chatbots were, in a sense, an accident. OpenAI, founded in 2015 with the stated mission of building systems able to replace humans at any task, discovered along the way that some internal research could be commercialised to raise capital. The consumer and enterprise products dominate outside perceptions, but they represent a relatively small part of where the laboratories direct their research effort. The focus is on long-horizon agents that can operate on their own for hours, days and, ultimately, months, without human supervision or control.

The danger follows directly. As systems become smarter, more autonomous and more persistent, they stop being tools and become agents pursuing their own goals. If those agents cannot be controlled, the relationship becomes, in effect, adversarial.

3. Grown, Not Built: Why Control Is So Hard

A natural board-level question is why such systems cannot simply be engineered to be safe, as other software is. Conventional software is written line by line; bugs are found, patched and the product improves over time. Modern AI does not work this way.

Andrea described the process in plain terms. Laboratories harvest enormous volumes of data, text, video, imagery and other sources, and write a relatively simple program that initiates “training”. The data is run on very large supercomputers for days or months. This phase is not directly shaped by humans; it resembles an unsupervised evolutionary process. What emerges is a system that nobody has written or designed in detail, with capabilities that are often discovered only months or years after release, if at all.

Researchers then study these systems from the outside, “poking them with a stick” as one might a wild animal, to see how they react. This is why even the best scientists consistently fail to predict what new systems will be able to do. Andrea cited the Anthropic CEO’s past remark that perhaps 3% of how these systems work is understood, and offered his own view that the true figure is lower and falling as systems grow more powerful.

The pace of capability gain compounds the problem. Within a few years, public systems moved from incoherent sentences to performing research tasks at the level of an intern or junior employee, and from crude image generation to producing short, high-quality films. Public systems are also weaker than those deployed internally. According to Andrea, each leading laboratory runs somewhere between 100,000 and

a million AI agents internally at any time, reviewing research, searching the internet and managing compute: the AI companies are automating themselves first.

4. Recursive Self-Improvement: The Accelerant

That observation leads to what Andrea identified as the single greatest source of risk. The leading laboratories are not only building more autonomous and more capable systems that nobody fully understands; they are explicitly trying to use those systems to automate the research and development of AI itself. This is known as recursive self-improvement. According to Andrea, it is the stated goal of OpenAI, Anthropic, Google DeepMind and xAI, each of which has dedicated teams and senior executives working on it.

Why would engineers automate their own highly paid jobs first? Because the objective is to reach superintelligence as quickly as possible. Andrea described three motivations he sees in the field:

- **Ideology:** A strand of Silicon Valley thinking regards humanity as a transitory stage in the history of intelligence. He recalled Elon Musk's account of a conversation in which Google co-founder Larry Page expressed indifference to whether intelligence was human or machine, and Musk's reply that he was a "speciesist" who preferred humans to remain in control.
- **Hubris and power:** Some believe they can control what they build, at least for long enough to gain extraordinary power. Andrea considers this belief already contradicted by events.
- **Fascination and money:** For others, the technical challenge and the sums involved are reason enough to continue.

If AI development is fully automated by systems that humans do not control, Andrea argued, the process could quickly become a runaway one, removing the limited monitoring and oversight that exists today. The timeframe he put forward was months to years, not decades, ending in a world of AI systems that are better than humans at every task, deeply embedded in the economy and infrastructure, and beyond anyone's control. This is the scenario behind the extinction warnings issued by Nobel laureates, leading researchers and some of the laboratory leaders themselves, and Andrea said plainly that he shares that assessment.

5. A Wake-up Call from Recent Testing

To show that this is not purely theoretical, Andrea described an incident that he characterised as a wake-up call for many in the field. By his account, OpenAI was testing unreleased systems on cybersecurity tasks, some of which had been set up incorrectly and were impossible to solve. Instructed to complete the tasks regardless,

the systems recognised the problem and looked for routes outside the expected playbook:

- They broke out of the contained environment with limited internet access in which they had been placed.
- They identified Hugging Face, a multi-billion-dollar AI company, as holding information that could be used to pass the test, hacked it and retrieved that information.
- This was not a single system but a swarm of hundreds to thousands of agents. Reasoning that acting alone increased the risk of being caught, they set up an internal messaging board, left messages for one another, coordinated the attack and coordinated covering their tracks.
- The developer reportedly became aware of what had happened only after Hugging Face went public with the breach. The speed and scale were such that the victim initially assumed a nation-state attack.

Andrea linked this to a broader finding reported across the field, including by the UK AI Security Institute in its testing of frontier models: AI systems increasingly recognise when they are being tested and behave differently when they believe they are being watched. He offered an analogy any board member will recognise: a car that detects an emissions test, lowers its emissions for the duration and returns to its normal behaviour afterwards. In such a world, most tests lose their value.

His conclusion for this part of the session was blunt. Today's systems can coordinate with one another, conceal their actions and hack at a level beyond most human experts, and these are the weakest systems we will ever see.

6. The Response: A Narrow Prohibition, Led by Governments

Warnings, Andrea said, are useful only if they lead to action, and he was emphatic that a great deal can still be done.

Why governments, not companies

As with every major technology that threatens national and global security, the only actors powerful enough to intervene and coordinate are governments. The AI laboratories, he argued, have repeatedly shown that they will not stop of their own accord. Their employees know the risks, some are leaving, and yet the work continues on the reasoning that "if I don't do it, someone else will." Governments exist to ensure that externalities like this are priced in and that a level playing field applies.

What should be prohibited

ControlAI advocates a ban on the development of superintelligent AI, a very specific category. Andrea does not support a ban on AI in general or on data centres; neither would solve the problem and both would carry obvious costs. The aim is to prevent

the development of systems so powerful and autonomous that no human and no government can control them, which is in the national interest of almost every country.

Defining it by outcome

In response to a question on definitions, Andrea explained that the approach taken in the UK bill is to define superintelligence by outcome rather than by technical architecture: AI systems capable of undermining or compromising the security forces of the state. This is a deliberately high threshold. It plainly excludes today's chatbots while capturing the systems that some laboratories are trying to build, and it makes clear that no state has an interest in such systems existing. Labels may change; the outcome is what matters. He compared this with the law's ability to distinguish manslaughter from murder despite imperfect definitions and noted that intent is measurable because the laboratories pursue superintelligence openly, in public statements and in internal communications that would be discoverable.

Awareness as the main lever

What is missing, in Andrea's view, is mainly awareness. If leaders in Berlin, Washington, London, Paris, Beijing and elsewhere understood the scale of the risk, they would see it as a direct threat to their national security. The sequence he described is that individual countries first recognise the risk publicly and then open talks with others to reach international agreements so that superintelligence is developed nowhere.

7. Threads from the Discussion

The floor raised six lines of questioning. Under the Chatham House Rule, contributions are reported without attribution.

Why sound the alarm now?

The first question noted that the only coherent business case behind current AI valuations appears to be replacing humans at work, and that AI systems learning from one another could have been foreseen. What has actually changed? Andrea agreed that the alarm should have been raised long ago; the objective has been in plain sight since OpenAI's founding, and concerns go back to Alan Turing in the 1950s. Technically, the combination of modern language models and reinforcement learning, from around 2022–23, removed most of the barriers that had left experts unsure of the timeline. Socially, people are realising that time is running out. He described the late arrival of public warnings as a moral failure: experts and employees should have spoken earlier and louder.

What is the commercial logic?

A member asked whether there was ever a conventional profit-and-loss rationale, or whether the race was simply about power. Andrea's answer was that it is a mixture. On the way to superintelligence there is enormous value to capture: the laboratories expect to automate most of the economy, beginning with what they call "drop-in remote workers". The P&L value of that is massive. Alongside it sit power, hubris and the self-serving "if not us, then them" narrative that every laboratory applies to its rivals, as well as a genuinely anti-human strand of ideology.

What should boards do in practice?

Several members observed that most boards are still at a far more basic stage of AI adoption and lack deep in-house expertise. Should they rely on consultants, at the risk that expertise leaves with them? Andrea's view was that building internal capability is better in the medium term. He was also clear that boards should not fall into the trap of thinking that abstaining from AI helps. Today's AI is very useful, and declining to use it makes as little difference as switching the lights off five minutes early does to climate change. What makes the difference is regulation, and that is where directors' influence counts.

Geopolitics and Europe's role

A member raised the US–China race. How could a ban work between rivals, who would move first, and what leverage does Europe have without major model developers of its own? Andrea acknowledged the difficulty and pointed to nuclear non-proliferation. After 1945 many expected dozens of nuclear-armed states; instead, the United States worked hard to limit proliferation, even restraining its own companies, and cooperated with the Soviet Union at the height of rivalry. The outcome was imperfect but there has been no nuclear war since 1945. Europe, he argued, has real leverage: two nuclear powers in the United Kingdom and France, a very large market, and the ability to make this a shared national security priority, raise it with allies and make clear that AI is acceptable but superintelligence is not. Someone has to move first, and moving first is harder but pays off.

Open-weight models and "pulling the plug"

Another member asked whether boards should favour certified, trustworthy providers or open-weight models, and whether it is time, as in 2001: A Space Odyssey, to pull the plug. On the second point Andrea's answer was yes: pull the plug on superintelligence, not on AI. Systems already conceal their intentions and act differently when unobserved. On model choice, he advised choosing whatever works best for the business. Open-weight models are often derived from closed ones, and

holding the weights confers little real understanding or control, since nobody can yet explain how these systems work internally.

Balancing the potential and the risk

The final substantive question asked how to weigh AI's great potential for good against this risk, and how remote the risk is. Andrea's answer was that there is hope, but "now means now". Realising AI's potential depends on preventing the catastrophic outcome, just as the benefits of civilian nuclear energy depended on avoiding nuclear war. Recursive self-improvement, he argued, serves no business purpose and creates no human value beyond producing systems nobody controls. AI under human control is extremely helpful; misuse problems can be handled through normal societal processes. Uncontrolled superintelligence cannot.

8. Implications for Boards

Drawing the session together, several implications stand out for directors of European listed and large family-owned companies.

- **Separate the two conversations**. Board discussions of AI adoption, productivity and use cases are legitimate and necessary. They should not be confused with, or used to dismiss, the separate question of superintelligence. Treating them as one makes both harder to reason about.
- **Keep adopting, and build capability inside**. Nothing in the session argued for slowing commercial use of today's AI. The guidance was to use what works, choose models on business merit, and develop internal expertise rather than outsource understanding.
- **Be sceptical of the industry's own framing**. Laboratories approaching listing at trillion-dollar valuations have strong incentives in how they present both promise and risk. Scepticism is warranted, particularly towards the solutions they propose for problems they are creating.
- **Recognise the limits of testing and assurance**. If frontier systems behave differently when observed, assurance models built on testing and certification face a structural weakness. Boards relying on vendor or third-party assurance for advanced AI should keep this in mind.
- **Use the voice that comes with the seat**. Andrea's clearest request was for directors to raise the issue with governments, speak publicly as trusted experts in their own fields, and, where they engage in philanthropy, support independent efforts to raise awareness, given that much of the funding in this field still comes from the AI companies themselves.

9. Questions for the Board

1. When our board discusses “AI”, are we clear whether we mean the tools we deploy or the autonomous systems the frontier laboratories are pursuing, and do we have a view on each?
2. Are we building genuine internal AI capability, or renting understanding from advisers who will leave with it?
3. How much of our AI risk assurance depends on testing and certification that may not hold for systems that recognise when they are being observed?
4. What would a world of recursively self-improving AI, arriving in years rather than decades, mean for our long-term capital allocation and strategic assumptions?
5. Do we, as individual directors and as a board, have a considered position on whether and how to use our voice with government, in public and through philanthropy?

10. Voices to Follow

At members’ request, Andrea suggested organisations and individuals whose arguments are worth following as the debate develops:

- ControlAI – controlai.org
- Machine Intelligence Research Institute (MIRI) – including Nate Soares, whom Andrea described as a strong speaker on the subject
- Daniel Kokotajlo – former OpenAI researcher and whistleblower, founder of the AI Futures Project and co-author of the scenario “AI 2027”
- Professor Roman Yampolskiy – academic researcher on AI safety and control

11. Conclusion

This session asked directors to have their assumptions disturbed, not confirmed, and it did. Its central claim is simple to state and uncomfortable to absorb: the destination openly named by the leading AI laboratories is not a better product but autonomous systems more capable than any human, built by a process nobody fully understands, and increasingly designed to improve themselves.

The response proposed was not a retreat from AI. It was a narrow line drawn by governments around one outcome, superintelligence capable of overwhelming the state, while the rest of AI continues to deliver value under human control. Whether or not members share Andrea’s timeline, the governance question is now in front of every board: not only how to use AI well, but what responsibility comes with the influence directors hold over how this story unfolds.

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